Università degli Studi di Napoli
Federico II
Dipartimento di Ingegneria Elettrica e delle Telecomunicazioni

Statistical Learning and Data Analysis – Module A

DATA PREPARATION

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More thoughts

"The available data are never ready for data analysis processing. They need to be Pre-Processed!"

"Success depends upon previous preparation, and without such preparation there is sure to be failure."—Confucius

GIGO (Garbage In – Garbage Out) – Information Technology

Data Excellence through Data Governance

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An example

- Assume that you have access to company's databases and data warehouse to identify features (item, price, units sold, shop type, distribution line, city, etc.) for your data analysis to perform your marketing task.
- **What do you discover?**
  - You notice that for several features there are no recorded value;
  - There can be some useful information missing, such as to know the promotional weeks, whether the item was purchased on sale or not;
  - There are unusual values, so that those who have recorded data made errors;
  - When you start pre-processing the data you find inconsistencies in the data and their coding.
- **Thus, the data you wish to analyze are:**
  - ***incomplete*** (missing values, lacking values of certain features of interest, or containing only aggregate data);
  - ***inaccurate* or *noisy*** (containing errors, or values that deviate from the expected);
  - ***inconsistent*** (e.g., containing discrepancies in the coding).
- **Welcome to the real world!**
  - *The raw format of real data is usually widely variable.*
  - *Many values may be missing, inconsistent across different data sources, and erroneous.*
  - *For the analyst, this leads to numerous challenges in using the data effectively.*

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Data need to be pre-processed!

In evaluating the interests of consumers from their activity on a social media site, the analyst may first need to determine the types of activity that are valuable to the mining process:

the interests entered by the user,
the comments entered by the user,
the set of friendships of the user along with their interests.

All these pieces of information are diverse and need to be collected from different databases within the social media site.

Some forms of data, such as raw logs, are often not directly usable because of their unstructured nature.

Useful features need to be extracted from these data sources.

Therefore, a data pre-processing and cleaning phase is needed.

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Practical Rule

"The available data are never ready for Data Processing".

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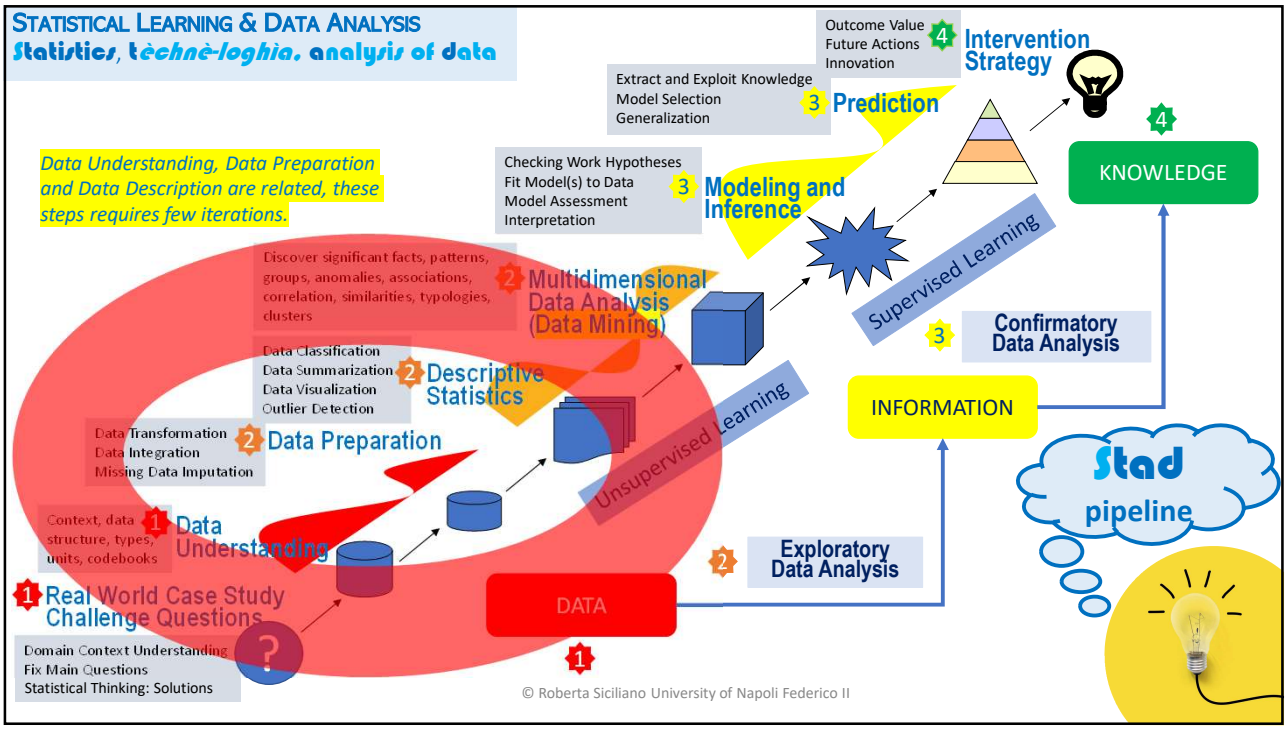

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Data Understanding, Data Preparation, Data Description

Data acquisition

- Data access
- Data integration
- Data collection report

Data description

- Variables/Features
- Units/Cases
- Descriptive statistics/Data Summarization
- Data description report/Summarization Report

Data quality assessment

- Missing values
- Outliers
- Data quality report

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How to describe data

- ✓ **Data Classification in Tables** (i.e., frequency distribution table, intensity distribution table)
- ✓ **Statistical Indices** (i.e., mean, variance, standard deviation, asymmetry, kurtosis)
- ✓ **Data Visualization** (i.e., bar chart, histogram, box-plot)

Data description is fundamental for Data Preparation (data pre-processing and cleaning) so that the two steps require some iterations.

Descriptive Statistics will be discussed in the next Lecture!

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Data quality

- Data have quality if they can be used, are useful and reliable.
- Key issues for Data Quality are:
  - ✓ ***Accuracy***
  - ✓ ***Completeness***
  - ✓ ***Consistency***
  - ✓ ***Timeliness***
  - ✓ ***Reliability***
  - ✓ ***Interpretability***

Data Preparation: Main Steps

Feature extraction and portability

- The raw data is often in a form that is not suitable for processing (i.e., raw logs, documents, semi-structured data, other forms of heterogeneous data).
- Some algorithms may work only with a specific data type, whereas the data may contain heterogeneous types (requiring portability).

Data cleaning

- Erroneous and inconsistent entries are removed, missing are estimated by imputation methods.

Data reduction, selection, and transformation

- The size of the data is reduced through data subset selection, feature subset selection, data transformation.
- Two gains:
  - ✓ data processing is more efficient;
  - ✓ removing irrelevant features, records, as well as redundant information the quality of the data mining process is improved.

Advanced Methods of Data Preparation are discussed in the Course of Data Mining!

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Basic issues in Data Preparation

Data Cleansing:
How do I clean up the data?

Data Transformation:
How do I express the variables?

Data Imputation:
How do I handle missing values?

Data Weighting:
Are all cases treated the same?

Data Filtering:
What do I do about outliers and other unwanted data?

Data Abstraction:
How do I handle temporal (time-series) data?

Data Reduction:
Can I reduce the amount of data to use?

Data Sampling: Records?

Dimensionality Reduction: Variables?

Data Discretization: Values?

Data Derivation:
Can I create some new variables?

Advanced Methods of Data Preparation are discussed in the Course of Data Mining and Machine Learning Module A!

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Data Transformation Types

- Normalization for scaling numerical values within a fixed range
- Discretization by replacing interval values/ordinal category to values
- Recoding nominal data to reduce the number of categories
- Feature construction by adding new features
- Aggregation to summarize more features
- Smoothing to remove noise from data

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Scaling and Normalization

- Features have different scales
  - Age versus Salary
- Standardization
- Min-max scaling
  - Map to [0,1]
  - Sensitive to noise

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Categorization and quantification

A numeric variable can also be *categorized* into distinct interval values yielding to an ordinal variable.

Some data transformation enable a categorical variable to become a numeric one. This will be the result of applying suitable methods of Data Mining.

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Numeric to Categorical Data: discretization

- Divides the range of the numeric variable into *k* intervals
  - ✓ *Equi-width intervals* (divide the range into *k* equal length sub-ranges).
  - ✓ *Equi-log intervals* (geometrically increasing, alike exponential distribution).
  - ✓ *Equi-depth intervals* (equal number of records, same granularity).
  - ✓ *Visualization and partitioning*
  - ✓ *Domain criterion*

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Categorical to Numeric Data: Binarization

Two categories

- 0,1 or -1,1

K categories

- *K*-dimensional indicator vector
- The position of 1 indicates the category

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Text to Numeric Data

Information Retrieval Processing

- Tokenization for data security
- Stop Word Removal
- Stemming reducing inflected words to their word base or root
- Weighting (Term Frequency - Proportional Document Frequency (TF-IDF)

Document-Term Matrix Processing

- Dimensionality Reduction through Latent Semantic Analysis (LSA) in Data Mining and Machine Learning Module A Course
- Normalization such that documents of varying length are treated in a uniform way

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Data Cleaning is necessary

Data collection technologies are inaccurate
- Sensors
- Scanning (optical character recognition is not perfect)
- Speech-to-text data

Privacy reasons
- Birthday incorrectly imputed in automated registration sites of social networks
- Several fields are empty

Manual Errors

Data collection is expansive
- Medical test

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Data Editing and Missing Imputation

- **Editing** (procedures)
  - review reported/keyed data for errors and pinpoints "inconsistent" values
- Items that fail edits are:
  - *referred to an analyst (manual Editing)*
  - *automatically **imputed** (replaced with consistent values)*

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Data Editing

- Data editing is the process whereby **survey data** are examined *for searching errors, in the form of missing, unusual or logically inconsistent values.*
- Data editing is a part of data-preprocessing: *it attempts to at least modify data, as little as possible, so as to render them complete and logically consistent for further data-processing.*
- Data editing requires **automatized processes** with the help of software to perform suitable methods for editing (i.e., clean-up), imputing (fill-in) missing or contradictory data, merging (fusion) data from different survey samples, such to enhance the ability of survey managers and analysts to obtain accurate estimates of micro-data to be further processed.

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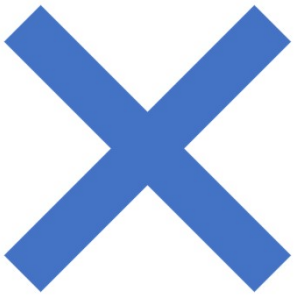
The Basics: What Is Imputation?

- **Imputation** is the replacement of a missing or incorrectly reported item using logical edits or statistical procedures.
- Imputation replaces a missing or incorrect data item with an **"educated guess."**



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Handling missing entries

Delete the data record containing missing entries

- What to do if nothing left?

Estimate or Impute the missing values

- Additional errors may be introduced
- Good under certain conditions (i.e., Matrix Completion)
- For dependency-oriented data, interpolation of behavioral attribute nearby values

Design an algorithm for missing data

- Missing Data Imputation Algorithms in Data Mining and Machine Learning Module A Course

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Handling incorrect and inconsistent entries

Inconsistency detection	When data is available from different sources in different formats • Data Integration within the database field *i.e., full name and abbreviation*
Domain knowledge	Ranges of the features or rules specifying the relationships across different features • Data scribbing and data auditing tools *i.e., age cannot be 800*
Data-centric methods	Use the statistical behavior to detect outliers • Examine before discarding

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SCAN ME

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